

Lab Report 7 – INCLINE PLANE & FRICTION

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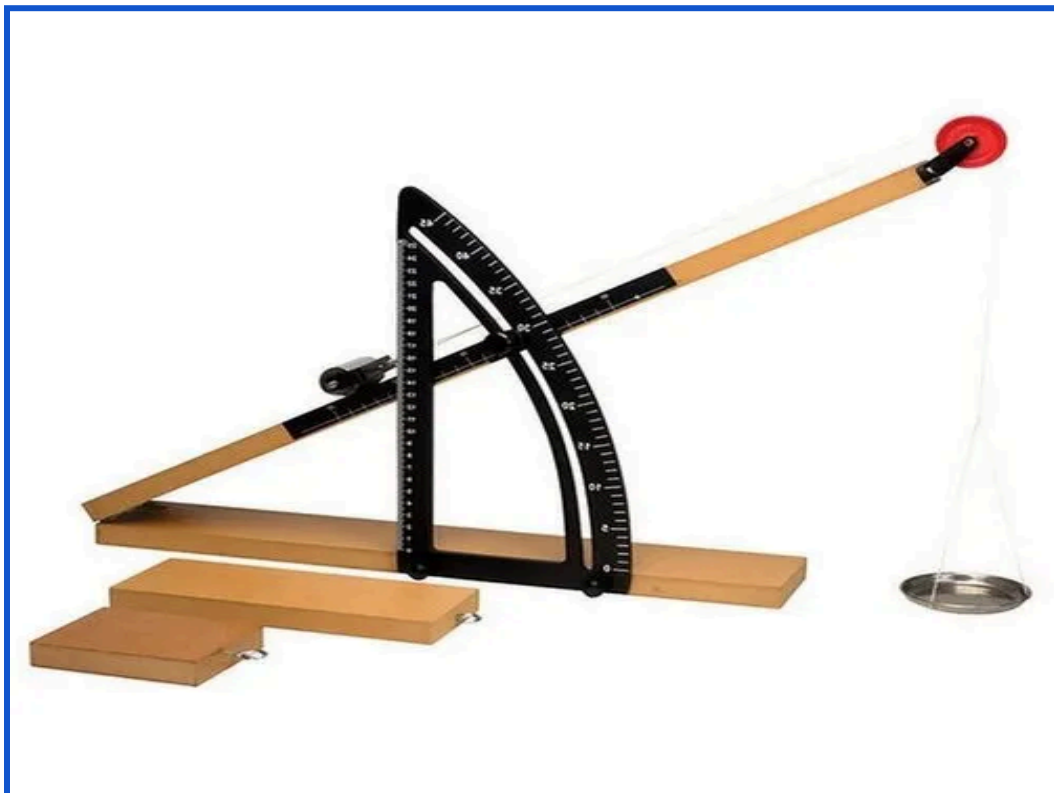
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PHYS 2125

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Objectives

To observe motion along an inclined plane and measure coefficients of friction.

Equipments

The list of equipments used in this experiment is:

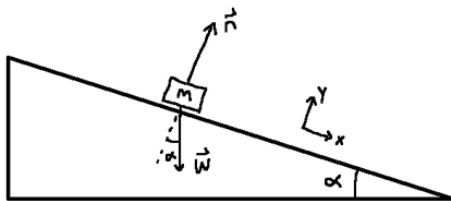
- Incline Plane
- Angle Indicator
- Motion Sensor
- Computer Interface
- Wheeled Cart
- Friction Carts

Theory and Equations

The motion of an object on an inclined plane involves 2 situations and related formulas: frictionless incline and kinetic friction.

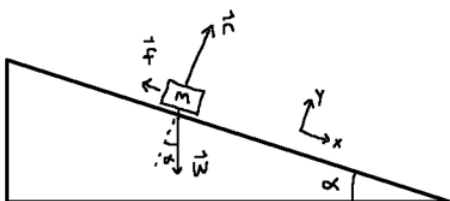
1. Frictionless Incline

In this situation, the object motion only is affected by the normal force and gravity:



$$g \sin \alpha = a_x \quad (\text{Eq. 1})$$

2. Kinetic Friction



$$\tan \alpha - \frac{a_x}{g \cos \alpha} = \mu_k \quad (\text{Eq. 2})$$

If the angle is too shallow, the mass will stay at rest, thanks to static friction:

$$f_s \leq \mu_s n$$

There will be some maximum angle that will allow the mass to remain at rest, but increasing the angle any further will cause the mass to accelerate. This occurs when static friction is at a maximum as well:

$$\tan \alpha_{max} = \mu_s \quad (\text{Eq. 3})$$

3. Percentage Error

The percentage error is calculated based on the experimentally determined measure (exp) and the accepted value (acc). And there are 2 types of accepted values which are theoretical value which is predicted from theory and empirical value which is collected from a reliable source.

$$\text{Percentage Error} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% \quad \text{units: \%}$$

Summary of Procedures

First, we set up a stable surface with the above equipment tools and inclined plane. Then, we set up the computer interface required to measure the friction force.

In part 1, we start the inclination angle at 2 degrees. We place the wheeled cart on the inclined track and hold it at rest. Next, we start recording on the computer interface, then release the car and try to catch it before it falls off the inclined plane. We use the computer interface to measure the acceleration of the cart. We increase the angle by 2 degrees and repeat the process until the angle reaches 10 degrees.

In part 2, we replace the wheeled cart with 3 other types of friction carts: cork bottom, plastic bottom, and felt bottom. First, we gradually increase the inclination of the plane to determine the maximum angle that can be achieved so that the cart remains stationary, then record that angle. Next, we set the angle slightly larger so that the cart always slides down the plane when released. We use the computer interface to record the acceleration of the motion. We continue to increase the angle by 2 degrees and repeat the process until we have a total of 5 acceleration values and their corresponding angle.

Data and Observations

Part 1

Inclined Plane	Wheeled car
Angle (degrees)	Actual Acceleration (m/s ²)
2	0.274
4	0.593
6	0.966
8	1.470
10	1.770

Part 2

Type of Car	Cork-bottom Car		Plastic-bottom Car		Felt-bottom Car	
Status	Angle (degrees)	Actual Acceleration (m/s ²)	Angle (degrees)	Actual Acceleration (m/s ²)	Angle (degrees)	Actual Acceleration (m/s ²)
(Static) Rest Angle	18		14		19	
Released Status	20	0.139	15	0.403	22	0.239
	22	0.314	17	0.701	24	0.648
	24	0.508	19	1.150	26	1.04
	26	0.828	21	1.710	28	1.38
	28	1.020	23	1.780	30	1.74

Data Analysis

Accepted g (m/s ²)	9.800
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Part 1:

Sample Calculation:

As the situation is frictionless incline, we apply the Eq. 1 to calculate the theoretical acceleration.

- Angle = 2 degrees

$$\text{Theoretical Acceleration} = a_x = g \sin \alpha = 9.800 \text{ m/s}^2 * \sin(2^\circ) = 0.342 \text{ m/s}^2$$

Inclined Plane	Wheeled car	
Angle (degrees)	Actual Acceleration (m/s^2)	Theoretical Acceleration (m/s^2)
2	0.274	0.342
4	0.593	0.684
6	0.966	1.02
8	1.470	1.36
10	1.770	1.70

Part 2:**Sample Calculation:**

Cork-bottom Car:

- Coefficient of Static Friction:

$$\mu_s = \tan \alpha_{max} = \tan(18^\circ) = 0.325$$

- Coefficient of Kinetic Friction at $\alpha = 20^\circ$

$$\mu_k = \tan \alpha - \frac{a_x}{g \cos \alpha} = \tan(20^\circ) - \frac{0.139}{9.800 * \cos(20^\circ)} = 0.349$$

- Average Kinetic Coefficient:

$$\text{Avg. Kinetic Coef.} = \frac{\sum \text{Coefficients of Released Status}}{\text{the amount of measurements}} = \frac{0.349 + 0.369 + 0.388 + 0.394 + 0.414}{5} = 0.383$$

Types of Car	Cork Bottom car			Plastic Bottom car			Felt Bottom car		
Features	Angle (degrees)	Actual Acceleration (m/s^2)	Coefficient	Angle (degrees)	Actual Acceleration (m/s^2)	Coefficient	Angle (degrees)	Actual Acceleration (m/s^2)	Coefficient
(Static) Rest Angle (degrees)	18		0.325	14		0.249	19		0.344
Released Angle Status (degrees)	20	0.139	0.349	15	0.403	0.225	22	0.239	0.378
	22	0.314	0.369	17	0.701	0.231	24	0.648	0.373
	24	0.508	0.388	19	1.150	0.220	26	1.04	0.370
	26	0.828	0.394	21	1.710	0.197	28	1.38	0.372
	28	1.020	0.414	23	1.780	0.227	30	1.74	0.372
Average Kinetic Coefficient (m/s^2)			0.383			0.220			0.373

Results

Part 1:

Sample Calculation:

- Angle = 2 degrees:

$$\text{Percentage Error} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% = \left| \frac{0.342 - 0.274}{0.342} \right| * 100\% = 19.9 \%$$

Inclined Plane	Wheeled car		Acceleration Comparison (Percentage Error)
Angle (degrees)	Actual Acceleration (m/s^2)	Theoretical Acceleration (m/s^2)	
2	0.274	0.342	19.9
4	0.593	0.684	13.3
6	0.966	1.02	5.70
8	1.470	1.36	7.78
10	1.770	1.70	4.01

Part 2:

Sample Calculation:

Cork-bottom Car:

- Percentage Error:

$$\text{Percentage Error} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% = \left| \frac{0.383 - 0.325}{0.325} \right| * 100\% = 17.8 \%$$

- Ranking Cars (based on Friction Coefficients:

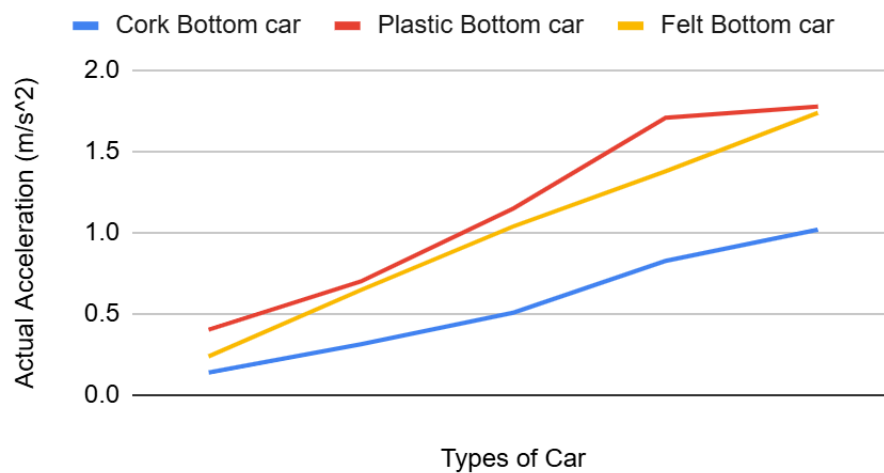
Highest car = Max(Average Kinetic Coefficient) = 0.383 => Cork-bottom Car

Lowest car = Min(Average Kinetic Coefficient) = 0.220 => Plastic-bottom Car

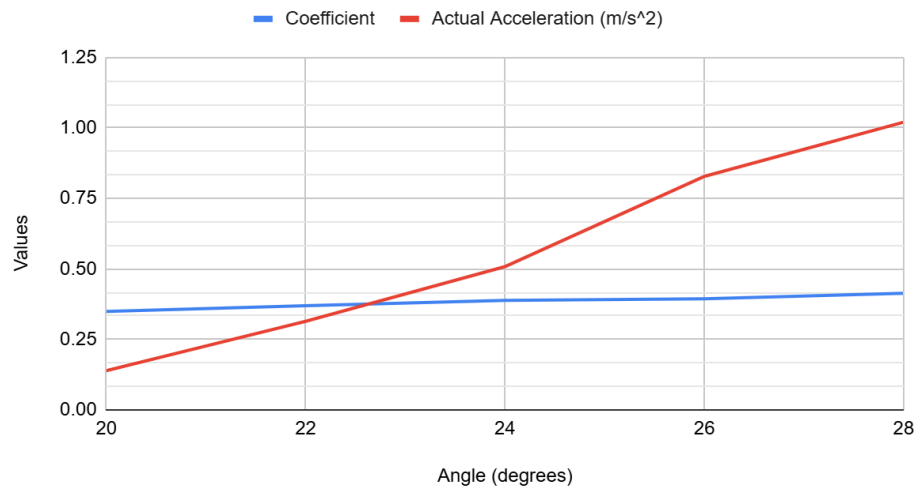
=> Medium car = Felt-bottom Car

Types of Car	Cork Bottom car			Plastic Bottom car			Felt Bottom car		
Features	Angle (degrees)	Actual Acceleration (m/s ²)	Coefficient	Angle (degrees)	Actual Acceleration (m/s ²)	Coefficient	Angle (degrees)	Actual Acceleration (m/s ²)	Coefficient
(Static) Rest Angle (degrees)	18		0.325	14		0.249	19		0.344
Released Angle Status (degrees)	20	0.139	0.349	15	0.403	0.225	22	0.239	0.378
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	26	0.828	0.394	21	1.710	0.197	28	1.38	0.372
	28	1.020	0.414	23	1.780	0.227	30	1.74	0.372
Average Kinetic Coefficient (m/s ²)			0.383			0.220			0.373
Percentage Error (%)			17.8			11.7			8.3
Ranking Car (based on Average Friction Coefficient) (m/s ²)	Highest	0.383	>>> Cork car						
	Medium	0.373	>>> Felt car						
	Lowest	0.220	>>> Plastic car						

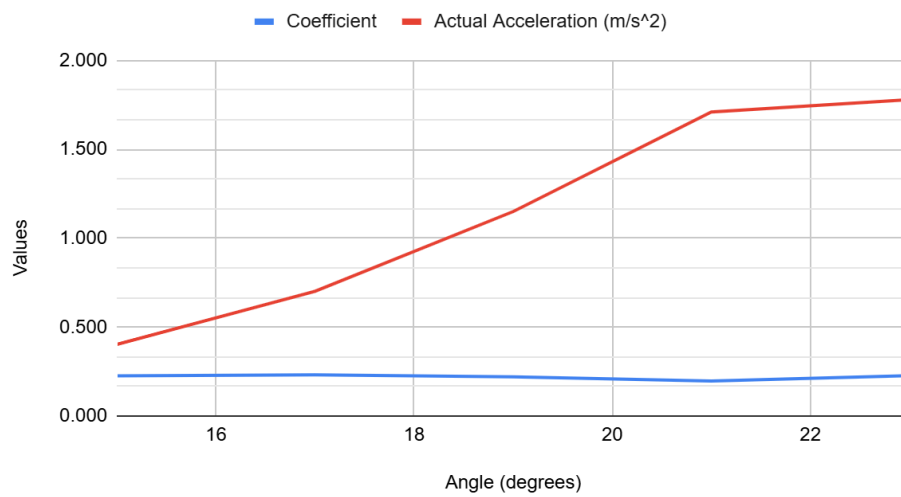
Actual Acceleration of Different Types of Car



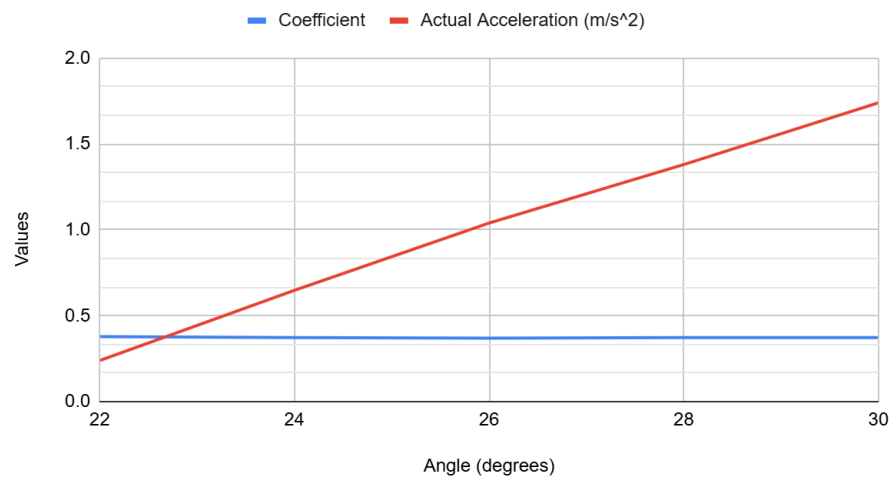
Measurements of Cork-bottom Car



Measurements of Plastic-bottom Car



Measurements of Felt-bottom Car



Conclusion

In the data table of part 1, we can see that as the angle of the inclined plane increases, both the actual acceleration and the theoretical acceleration of the car increase. This totally fits Equation 1. Since the inclined car lies within the domain of (0 degrees, 90 degrees) in Quadrant 1 of the unit circle, sine is positive and increases as the angle increases. Furthermore, we also see that as the angle increases, the percentage error decreases. This means that the theoretical measurement of acceleration for the frictionless incline case more closely approximates the actual measurement of the car's acceleration at large angles.

In part 2, from the graphs of “Actual Acceleration of Different Types of Car”, we can confirm again that even if there is a friction force, the acceleration still increases when the angle increases. Moreover, in the 3 graphs of measurements of each car, we can see that the coefficient in each case is almost constant when the angle increases. This confirms that there exists a constant of coefficient of kinetic friction coefficient as the Eq. 2 mentions for a given object (car type). Furthermore, depending on the different types of car, the friction force is different. For some materials like cork, it produces more friction force than other materials, that is why the cork-bottom car has the highest friction coefficient in comparison to others.